CS19541 – COMPUTER NETWORKS LAB

EXPERIMENT – 9

WIRELESS ROUTER, DHCP SERVER AND INTERNETWORK CONFIGURATION

1. Key Features of Wireless Router and DHCP Server

Wireless Router: A wireless router enables devices to connect to a network without cables. Its settings include the wireless network name (SSID), for example, HomeNetwork. It may also provide DHCP service and use a specified DNS server address.

DHCP Server: A Dynamic Host Configuration Protocol (DHCP) server automatically provides devices with essential network settings such as an IP address, subnet mask, default gateway and DNS server address. This avoids configuring each device manually.

2. Importance of DHCP in Internetworking

DHCP simplifies network setup by automatically assigning the required configuration to connected devices. It reduces manual work, helps avoid IP address conflicts, and makes network administration easier. When a device joins the network, it can receive its network settings automatically.

3. Design and Configuration of an Internetwork

Create the network in the lab by connecting the devices in this order:

`PC0 → Switch → Router → Switch → PC1`

Use copper straight-through Ethernet cables to connect the PCs, switches and router interfaces.

IP Address Configuration

Device / Interface	IP Address	Subnet Mask	Default Gateway
Router Fa0/0	192.168.10.1	255.255.255.0	—
PC0	192.168.10.2	255.255.255.0	192.168.10.1
Router Fa0/1	192.168.20.1	255.255.255.0	—
PC1	192.168.20.2	255.255.255.0	192.168.20.1

Connectivity Verification

Check communication between PC0 and PC1 using the **ping** command. In Cisco Packet Tracer, you can also use the **Simple PDU** tool to send a test packet from PC0 to PC1.

Result

Thus, the key features of a wireless router and DHCP server were reviewed, and an internetwork using PCs, switches, a router and Ethernet cables was designed with the given IP addressing scheme.

Note

The uploaded source labels these questions under Practical 10; this record keeps the requested Experiment 9 title.